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MODULE-2

TOOLS FOR DESIGN THINKING

Real-Time design interaction capture and analysis – Enabling efficient collaboration in digital space– Empathy for design – Collaboration in distributed Design

Design thinking tools, methods, practices, and frameworks are part of both visible and invisible work, which is always being done “backstage” as part of most companies’ innovation processes. More specifically, it is the heart of the process of building environments (physical and now virtual) that help facilitate new business ideas.

We have recently released our Innovation Report, where we shared our belief that Design Thinking still makes a difference within the business innovation context. Furthermore, we think it will continue to be impactful throughout the next decade. However, we don’t want to leave you with such a strong statement without giving you a taste of something more tactical. Something you can put into practice now.

We are not saying that you should run an entire Design Thinking process or an expert holistic diagnosis by yourself. You, of course, will need professional help to unlock such opportunities. But by using the following tools, you will become aware of the new possibilities and potential that design thinking and its processes can offer your teams; how the same co-workers will be able to think differently and present new and more consistent and structured ideas; and how their implementation can leverage the level of your deliveries.

The time for chit-chat is over, the moment has arrived for you to check out our 30-tool-starter-pack to boost your innovation.

What is Design Thinking?

When we talk about Design Thinking, we are referring primarily to the way designers think. They use unconventional reasoning in the business world, abstract thinking to:

- Formulate questions through the assumption or understanding of the current situation or scenario.
- Elaborate questions from information collected during the observation of the universe that surrounds a problem.
- Create solutions that derive from the problem or specific pain-points.

Innovative Solutions

We can define Design Thinking as: “A structured, human-centered approach to innovation which seeks to generate solutions that align user’s desires and needs with business value generation.”

To make it even simpler, you could say that Design Thinking is an innovative approach based on the designer's thoughts, combining creativity and empathy to create innovative solutions.

What are some Design Thinking tools?

To list the Design Thinking tools, we need to remember that it is a process generally divided into four phases:

- **Immersion:** research to contextualize the problem.
- **Analysis and Synthesis:** grouping collected data and reframing the initial situation based on transforming data into insights.
- **Ideation:** collaborative brainstorming sessions with the help of tools to create innovative solutions.
- **Prototyping:** testing to validate the effectiveness of the solutions at bringing value to the end-user.

IMMERSION TOOLS

1. Exploratory Research

A preliminary field research for the team to understand the context surrounding the problem.

2. Desk Research

Search for information on the project's theme from different sources: websites, books, magazines, blogs, articles, etc.

3. In-depth Interviews

Obtaining information through dialogue, mainly with users/developers about the product/service/process.

4. Awareness Notebooks

Instruments used to obtain data, usually when the user is physically distant.

5. Ethnographic research

Ethnography is a branch of anthropology, which aims to understand behaviors and cultural relationship dynamics by conducting field research. It achieves its objectives by talking with and observing people and their social interactions. It can be used with groups, teams, organizations, in a nutshell, with every kind of group.

It's a qualitative method where researchers observe and/or interact with a study's participants in their real-life environment with no preparation — doing wherever they are, doing whatever they do.

The concept was made famous in anthropology, but can be used across a wide range of social sciences, including marketing and digital disciplines, such as user experience.

6. A Day in the Life

Simulation of a person's life or specific situation.

7. Generation Research

Meetings with team members and stakeholders to carry out activities to present their views and share their experience with the project thus far.

8. Focus Group

Survey used to check people's reactions to a particular issue or product. The method is considered an advanced qualitative research technique and should be used when one wants to understand consumer habits, observing their particularities and individual behavior.

It is conducted by a mediator and carried out by volunteers who meet in-person to answer open-ended questions about a specific and predetermined topic.

9. Shadowing

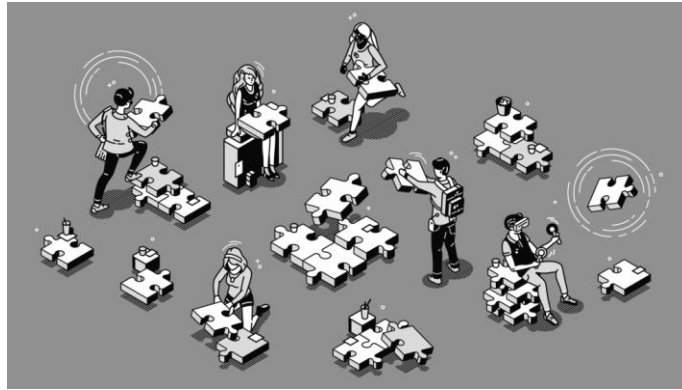
Monitoring a user over a certain period that includes their interaction with the product or service under analysis.

ANALYSIS AND SYNTHESIS TOOLS

10. Insight Cards

Reflections based on real data from Exploratory, Desk, and In-Depth surveys, transformed into cards that facilitate the visualization of information.

11. Affinity Diagram



The Affinity Diagram is a tool that allows you to organize ideas during brainstorming sessions. The goal is to take large amounts of information and/or insights and understand the essence behind that content.

Essentially, the proposal is to group ideas based on affinity, similarity, dependence, or proximity. Then put them into a diagram within the macro areas that identify a topic to be worked on, subdivisions, and interdependencies. Get closer to your target's interests using affinity diagrams.

12. Concept Map

A Concept Map is a simplified diagram or visual organization of complex field data, at different levels of depth. It represents how two or more concepts/ideas are related to each other, enabling more linear reasoning and allowing new insights to be extracted from the information.

It's usually portrayed with nodes (boxes or circles), which are hierarchically structured and connected by the arcs (lines or arrows).

The idea behind the process is to create a knowledge system for any topic. In our recent experience working 100% remotely, some online tools have emerged as a solution for us to share these insights, such as Miro and MindMeister.

13. Guiding Criteria

Guidelines that must be followed continuously during the development of a project, which determine the limits of tasks, maintaining the proposed focus.

14. Empathy Map



The Empathy Map is a visual tool that analyzes and describes behavioral aspects of the ideal customer. With this simple and didactic Design Thinking resource, it is possible to detail scenarios, thoughts, actions, problems, and the needs of your target audience.

The more you know about your audience, the more you'll be able to understand what they need and how to help them fulfill their desires, problems, and expectations.

When there is a lot of field information, it is used to better concentrate on understanding the target audience.

The protagonist of the Map is always the customer, but it is also of use to app users, audience members, service consumers. In other words, anyone who relates to what your business offers.

15. Personas

Personas are fictional archetypes that embody the brand's values and represent the ideal customer's perspective.

If you live on this round piece of land called Earth, you're probably tired of hearing that personas are fictional and general representations of a target audience with similar attitudes, objectives, needs, and behaviors.

What may be slipping out of your field of vision is that personas play a relevant role in the customization of solutions, providing insights that can be used to promote personalized experiences.

16. User journey

A graphical representation of the stages of users' relationship with the product or service.

17. Blueprints



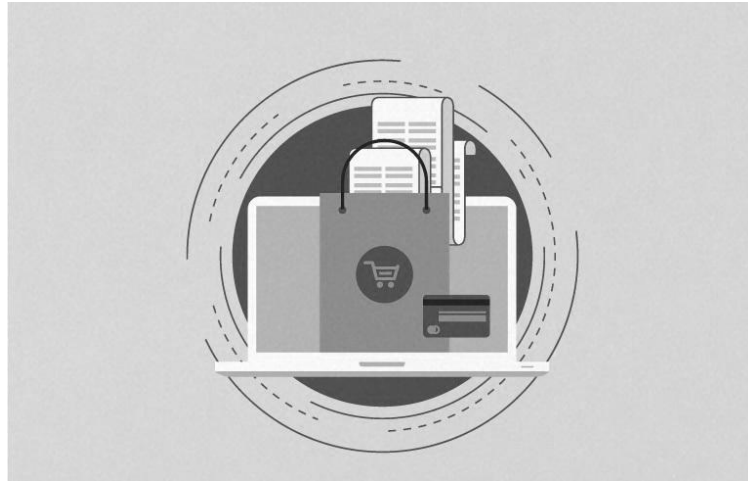
A service Service Blueprint is essentially a diagram that shows, in general, the relationships between different solutions (like products and services) and its components (people, physical or digital evidence, and processes), that are directly tied.

It works as a visual schematic matrix representing the whole system of interactions that straightforwardly characterize a service. Since blueprinting acts as a sort of magnifying glass for the customer journey, the best place to use it is within highly complex experiences. Services that take place over multiple touchpoints or require coordination between various departments.

18. Reframing

Examining unanswered questions in a company from different perspectives, allowing for the deconstruction of biases and assumptions about a business, product, or service.

19. Customer Journey Maps



A Customer Journey Map is a visual representation of every experience your customers have with you, your services, products, basically your entire brand. Journey maps illustrate all the touchpoints that your clients may have through visuals that tell the story of how they moved through each phase of interaction.

They are often based on a timeline of events, for example, from initial attraction – when a customer first makes contact with you, all the way through to their final purchase and post-purchase support.

IDEATION TOOLS

20. Brainstorming & Brainwriting

It is a creative process to encourage those involved in the project to generate many ideas quickly. Brainstorming meetings where before the creative discussion starts, everyone writes their ideas anonymously on pieces of paper, which are shuffled afterward.

21. Co-creation Workshop

collaborative meeting held by the Design team, which brings together individuals from different areas to foster innovative solutions.

22. Ideas Menu

Catalog that summarizes and makes tangible all the ideas generated in the project.

23. Positioning Matrix

A matrix that communicates the benefits and challenges in implementing each solution. This way, the most strategic ideas are prioritized for prototyping.

PROTOTYPING TOOLS

24. Proof of Concept

A Proof of Concept is a realization of a certain method or idea in order to demonstrate its feasibility. It's how we determine whether an idea can be turned into a reality. It is where we discover what actually works about an idea? And whether it will function as envisioned?

Developing a proof of concept generally requires some investment of time and resources. But by going through this process companies are able to determine a concept's viability before putting production-level resources behind an untested idea and potentially wasting valuable time, resources, and investment.

A POC is often the vital final cog in the engine that can be viewed as a deciding factor before a product is developed and launched.

25. Minimum Viable Product



A Minimum Viable Product, as you are probably aware, is the simplest version of a product, service, or functionality to obtain your value proposition's market validation.

It can help you avoid risks and provides a platform for product validation. It also separates ideas from execution, theory from practice, and the abstract from the concrete.

26. Volume Model

A three-dimensional representation of a product with varying fidelity levels is used to take the idea and transform it into something concrete.

27. Wireframing

Wireframing is a prototype used in interface design to sketch the structure of a digital product such as websites or applications, briefly exemplifying the relationships between its pages and other key elements in the interface.

In detail, it consists of a simple visual representation, usually in grayscale, of the structure and functionality of a single web page or a sequence of linked pages.

Wireframes can be sketched manually or digitally as long as it fulfills the purpose of structuring and validating ideas graphically.

28. Storyboard

Visual representations of a story through static frames. They are created from drawings, collages, photographs, or any other type of graphic representation.

29. Staging & service prototype

Staging is the name we call an improvised simulation of material artifacts, environments or even people's interactions with objects or dialogues. It is used to represent aspects of a solution, as well as test, build, or detail steps in a procedure to improve a product or service experience.

30. Prototyping on paper

Simple representations of interfaces, drawn by hand, with different fidelity levels to make an idea tangible.

Four steps to highlight the benefits of the Design Thinking tools

Design Thinking is not something that can be implemented overnight without specialized help. This does not mean that it is impossible to test it on a smaller project or work on changing your employees' mindset.

Below are some basic steps that must be taken to implement this process!

1. Focus on the problem

Companies often fail to solve problems effectively or meet goals because they do not correctly identify the problem from the beginning. Here are some tips for identifying your problem:

- Listen: put yourself in the shoes of your users and see through their lenses.
- Ask: who encounters this problem, and why? Why have previous attempts failed to resolve the issue at hand?
- Have collaborative conversations: working in cubicles is an easy trap to fall into. Interact with everyone, not just your team members.
- Stay impartial: don't assume that you immediately understand the problem or the solution. With an open mind, you can find something that you didn't expect.

2. Develop Design Thinking skills within your team

Traditionally, the Design Thinking process's design phase was performed by project managers or engineers. But that does not mean that only one department or function can do it.

Since Design Thinking is the act of asking questions, understanding, and testing, everyone can and should participate in it.

Here are some tips for developing your team's design skills:

- Practice the DT mindset: start implementing the process in your role whenever you can. For example, if you oversee integration, think of ways to test a new approach or understand your employees' mindsets by collecting feedback through a survey. Stay open to new results.
- Promote interests in Design Thinking: If you have team members who want to take the initiative and expand their skill sets, encourage that interest and experimentation within the company.

3. Ask more questions

It is essential to understand that Design Thinking is continuous. It is an iterative process that continually revisits topics and projects since there is always room for improvement.

However, learning can't happen if there is no feedback process. Here are some tips for creating a learning culture by collecting feedback:

- Be open about what went wrong: set an example by demonstrating that failure is an expected part of DT. Openly discuss which tests failed and why.
- See failure as learning opportunities: trying and failing in a new approach serves the crucial function of narrowing down the list of possible solutions. This brings you and your team closer to the method that works best. Encourage failure!

4. Embrace the feedback loop

The goal of Design Thinking is not perfection but the best possible answer. And the best solution is probably not the first. Thus, a constant feedback loop is essential.

Here are some tips for implementing feedback loops:

- Test and repeat as much as possible: find new ways and angles to test your assumptions. You'll find something you would never have otherwise.
- Conduct feedback sessions frequently: when you adopt feedback, it not only creates a safe space to innovate, but it also prevents the same mistakes from happening again.

And then? After you have a taste of design innovation, with the results proven, if you need consistent results and scale, you know what to do. [Reach us here](#).

As you have seen, Design Thinking is a very useful approach to solving complex problems. There is a reason why the most innovative companies have teams dedicated to it and invest in hiring specialized consultants on the subject.

REAL-TIME DESIGN INTERACTION CAPTURE AND ANALYSIS

What if we could capture knowledge and interactions in real time, in order to better understand team interaction and workflow?



WHAT WE SET OUT TO DO

We set out to develop a knowledge capture and reuse system that enables real-time analysis and feedback for design workspaces. Our aim was to leverage emerging sensor, microphone and video technology to develop a system that documents the entire creative and collaborative design process, enabling designers to review and react to their processes in real-time.

Researchers instrumented and observed project teams in action in ME310, Stanford's team-based design innovation course and in the Center for Design Research's (CDR's) Design Observatory. Based on initial findings, researchers developed and tested prototypes and interface concepts for real time and near-real-time design activity analysis and feedback.

WHAT WE FOUND

Researchers identified three design process needs, based on accessing past knowledge, enhancing current interactions and clarifying future directions.

To connect teams with the past knowledge, researchers built a searchable database of 12 years of ME310 team documents and assignments.

To improve teams' knowledge capture and interpersonal dynamics, researchers explored real-time conversation transcription, speaker identification, and sentiment analysis software. Researchers tracked speaker activity and intensity with a heat map of sound energy.

To improve teams' focus on future actions, researchers used transcripts and Wizard-of-Oz prototyping to identify key concepts and action items.

Generating a list of tasks-to-perform received positive feedback from students. Automating this process also proved to be a challenge, and was instead performed by a human coder reviewing meeting transcripts. Researchers applied conversation analysis such as Linguistic Inquiry and Word Count to identify conversation themes and sentiment. Finally, they also prototyped a critical moments interface, which allows teams to highlight moments that seem in-the-moment to be important, which can be used for other team

members to catch up quickly, or to allow teams to tag and later re-find those conversations during overall project design documentation.

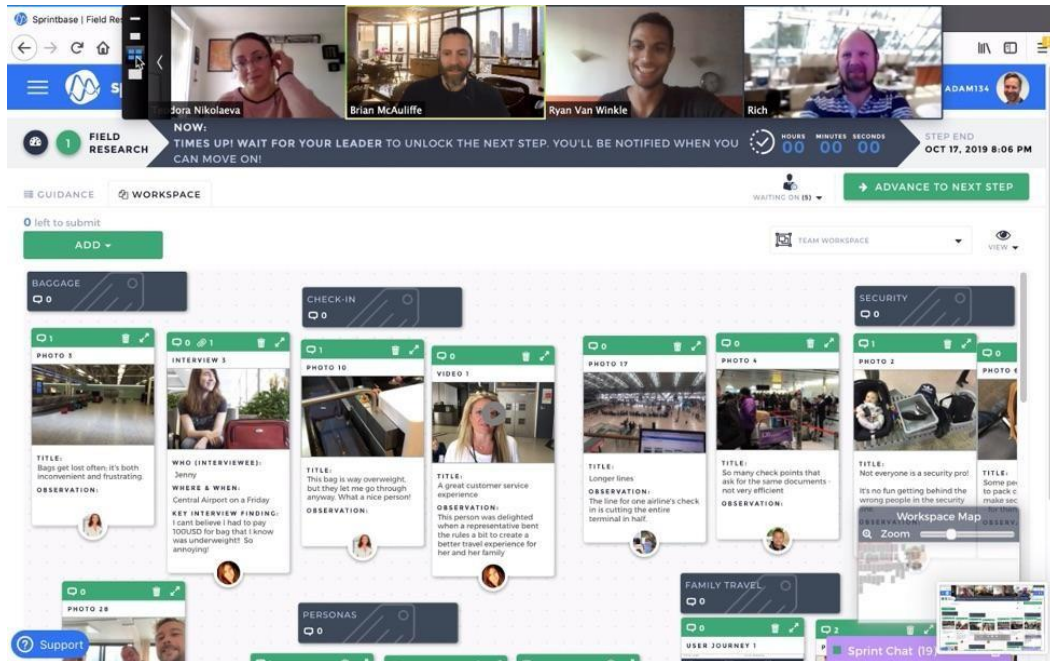
ENABLING EFFICIENT COLLABORATION IN DIGITAL SPACE

Design thinking is a human-centered approach to innovation, focused on gaining empathy about customer's problems and challenges in order to create solutions or products and services that satisfy their wants and needs. This framework or process requires different techniques and tools than your overall "business as usual" and product design efforts. To help with this approach, there are many software tools and applications that you can utilize during the design thinking process. Here are some popular software tools for design thinking that you can evaluate yourself to see if digital tools can help you.

1. Sprintbase
2. Miro
3. MURAL
4. Shape by IDEO
5. Smaply
6. Digsite
7. Batterii
8. Stormboard
9. Google Docs, Sheets, & Slides
10. Conceptboard
11. Google Jamboard
12. Shape
13. FigJam

1. SPRINTBASE

Sprintbase is a design thinking software program that guides teams through the innovation process step-by-step. The methods and tools help teams tackle their creative problems, learn to collaborate successfully, and save time and money in the process through engaging digitally. Sprintbase was developed by expert design thinking practitioners, has been featured in Forbes, and is used by organizations like Ebay, CapGemini, Deloitte, and ABInBev. Sprint Base helps remote teams confidently apply design thinking, embed innovation skills, and get results.



2. MIRO (Realtime Board)

Miro (formerly Realtime Board) is a simple online whiteboard tool for collaboration and creation. You can brainstorm new ideas together in real time, make connections between ideas and product solutions, and more. This is a great tool for the ideation and strategic planning process. We also like the visual nature of creating maps and diagrams to bring your ideas to life in a digital space. See our post to learn more about what is Miro and how to use Miro for virtual collaboration and realtime whiteboarding.

3. MURAL

MURAL is a visual collaboration space for your teams to work more efficiently. The platform provides shared, digital “whiteboards” that allows you to explore challenges and organize your ideas in one place. The features and tools they offer are growing rapidly and fit with many different frameworks and design thinking stages. Mural has a number of templates that you can work from and collaborate with others on that are based on some of the most popular activities in the design thinking, agile, and innovation world. It’s a good solution for remote collaboration. See our post to learn more about what is Mural and how to use Mural for design thinking and design sprints.

4. SHAPE (by IDEO)

Shape is the newest design thinking software on our list. It comes from design thinking influencers IDEO. Shape is a visual, collaborative space to build, test, and refine your ideas. Shape allows you to do many of the key activities of design thinking online.

- Gather inspiration and ideas.
- Engage and guide teams through proven processes.
- Activate communities and teams around shared strategic challenges.
- Quickly and flexibly gather customer feedback.
- Build and share a knowledge base.

5. SMAPLY

Smaply provides several unique editors that relate to design thinking processes. The first is a persona editor tool that allows you to create custom user personas based on your customers. The software also has a journey map editor and a stakeholder map designer so that you can collaborate with your team and design strategically from the very beginning of the process.

6. DIGSITE

Digsite is a platform designed for qualitative research to empower your ideas. Get to know your target customers intimately and quickly with their customized research focus-group-like methods.

7. BATTERII

Batterii is a multi-platform tool in which you can gain powerful insights into your customers. The software allows you to collect consumer insight research from anywhere on the web. You can keep everything together in one place, and manage your team's efforts easily with the tool.

8. STORMBOARD

Stormboard is a shared sticky note and whiteboard software tool for innovation teams. You can generate many ideas, prioritize them, organize, and refine them within this efficient and easy to use tool.

9. GOOGLE DOCS, SHEETS, & SLIDES

Before investing in a design thinking software solution many organizations will validate and prototype the idea with Google docs or whatever collaborative document solutions their organization provides. This is simply creating a google sheet, slide, or doc and inviting people to collaborate on it at the same time or over a longer time period. This helps people play with doing some design thinking activities remotely and collaboratively. They can be at any stage of the process such as summarizing interview findings

and defining a challenge with google docs, idea generation, selection, and rating with google sheets – or even creating a concept sheet or digital prototype with a google slide. We walk through an activity you can do with Google Sheets in our design thinking facilitator training program. After they see this has promise they may select an application more specifically designed for design thinking activities.

10. CONCEPTBOARD

Conceptboard is another virtual whiteboard and collaboration software tool for innovation-focused teams. With an infinitely generating blank canvas, you can work together with your team to generate ideas, organize thoughts, and quickly and easily narrow in on your focus and plans.

11. GOOGLE JAMBOARD

Google Jamboard is a very simple online realtime whiteboarding tool that is gaining in popularity. It offers a big upgrade in features to the whiteboards you find built into your online meeting tools (like Zoom's whiteboard feature). It doesn't have as many features as the popular Miro and Mural whiteboards. Google Jamboard is popular with teachers and in education, as well as with facilitators leading design thinking sessions online with groups that may not be very tech savvy. Google Jamboard does include the digital sticky notes and an image search tool that make collaborative whiteboards a powerful visual tool.

12. SHAPE

Shape is similar to many other whiteboard tools on this list, including Mural and Miro, in that it offers a visual blank canvas for users to collaborate and brainstorm together. This tool, developed by IDEO, also offers surveys, shared workspaces, and other helpful features like adding images, videos, and links within the canvas. Templates also make it easy to get up and running with the software quickly.

13. FIGJAM

FigJam by Figma is a robust whiteboard tool that allows you to use sticky notes and shapes, freehand draw, react with stickers and stamps, and copy and paste between the tool and the original Figma platform.

EMPATHY FOR DESIGN

Empathy is an important element in Design Thinking and Human-Centered Design. What is empathy exactly? Why is empathy so important to designing solutions that actually work for people? Here, we'll not only look at what empathy means, but will also look at how it helps design thinkers create solutions that work and, conversely, how a lack of empathy can result in product failure. We'll also come to understand the empowering notion that everyone can master empathy and design truly human-centered solutions.

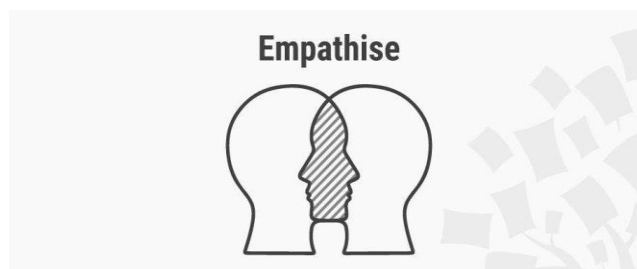
What Is Empathy Exactly?

In a general sense, empathy is our ability to see the world through other people's eyes, to see what they see, feel what they feel, and experience things as they do. Of course, none of us can fully experience things the way someone else does, but we can attempt to get as close as possible, and we do this by putting aside our own preconceived ideas and choosing to understand the ideas, thoughts, and needs of others instead.

In Design Thinking, empathy is, as explained in IDEO's Human-Centered Design Toolkit, a "deep understanding of the problems and realities of the people you are designing for". It involves learning about the difficulties people face, as well as uncovering their latent needs and desires in order to explain their behaviors. To do so, we need to have an understanding of the people's environment, as well as their roles in and interactions with their environment.

Empathy helps us gain a deeper appreciation and understanding of people's emotional and physical needs, and the way they see, understand, and interact with the world around them. It will also help us to understand how all of this has an impact on their lives generally, specifically within the contexts being investigated. Unlike traditional marketing research, empathic research is not concerned with facts about people (such as their weight or the amount of food they eat), but more about their motivations and thoughts (for instance, why they prefer to sit at home watching TV as opposed to going out for a jog). It's inherently subjective, since there is a fair amount of interpretation involved in finding out what people mean rather than what they say.

EMPATHISE



"Empathise" is the first stage of the Design Thinking process. The following stages can be summarised as: Define, Ideate, Prototype and Test. In the empathise stage, your goal, as a designer, is to gain an empathic understanding of the people you're designing for and the problem you are trying to solve. This process involves observing, engaging, and empathising with the people you are designing for in order to

understand their experiences and motivations, as well as immersing yourself in their physical environment in order to have a deeper personal understanding of the issues, needs and challenges involved.

Empathy is crucial to a human-centred design process such as Design Thinking, and empathy helps design thinkers to set aside his or her own assumptions about the world in order to gain insight into their users and their needs. Depending on your time constraints, you will want to gather a substantial amount of information at this stage of the Design Thinking process. In the Empathise stage of a Design Thinking process, you will develop the empathy, understandings, experiences, insights and observations on which you will use to build the rest of your design project. We cannot stress enough how important it is for designers such as us to develop the best possible understanding of our users, their needs, and the problems that underlie the development of the particular product or service we're aiming to design. If you have time and money, you should also consider consulting experts in order to find out more about the people you design for, but you'll be surprised at how much insight you and your team can easily gain via practical Empathise methods.

EMPATHISE METHODS

The following are our favourite Empathise methods:

- Assume a beginner's mindset
- Ask What-How-Why
- Ask the [5 whys](#)
- Conduct interviews with empathy
- Build empathy with [analogies](#)
- Use photo and video user-based studies
- Use personal photo and video journals
- Engage with extreme users
- [Story share-and-capture](#)
- [Bodystorm](#)
- Create journey maps

However, you will need to understand the following nuances and potentials of empathy before you start using the above (amazing) methods.

EMPATHY VS. SYMPATHY

Sympathy, a word often confused with empathy, is more about one's ability to have or show concern for the wellbeing of another, whereas to sympathize does not necessarily require one to experience in a deep way what others experience. Additionally, sympathy often involves a sense of detachment and superiority; when we sympathize, we tend to project feelings of pity and sorrow for another person.

This feeling of pity and sorrow may not only rub people up the wrong way, but it is also useless in a Design Thinking process. In Design Thinking, we are concerned with understanding the people for whom we are designing solutions—for doing something that can help them. When we visit our users in their natural environments in order to learn about how they behave, or when we conduct interviews with them, we are not

seeking for opportunities to react to the people; rather, we want to absorb what they are going through, and feel what they are feeling.

WHY EMPATHY?

Moving Away from the Industrial Revolution

Since the invention of factories in the industrial revolution opened the gates to mass-produced goods, mass consumerism has been an ever-growing part of how the world operates. However, the one-size-fits-all approach to consumption and solving problems has begun to show signs of inadequacy.

The truth is, using the power of “averages” is a terrible way to design solutions for people. In the 1940s, the US Air Force learnt this the hard way. During this era, aviation accidents happened very frequently (as many as 17 crashes a day). Initially, the air force presumed that the reason for so many accidents was the air force’s switch to using more complicated and faster planes. After some research, however, the air force discovered the real reason behind the accidents; they had designed the planes’ cockpits and helmets to conform to the dimensions of the “average” soldier’s body. In a study of over 4,000 air force pilots, it was found that none of the air force pilots fell within the dimensions of the supposed “average” man. It was no wonder pilots had problems with using the planes! In the end, the air force created adjustable equipment to fit most soldiers’ bodies, thereby solving the problem.

Besides the problems with designing solutions based on averages, our mass consumerism has a further issue: the high rate at which we are generating waste. In the past decade, our consumption has turned global warming from a growing issue to an imminent crisis that threatens to change the way we live (and even survive). Design Thinking, and in particular empathy, is about creating solutions that are sustainable and focused on all pertinent areas that can affect us in the long term.



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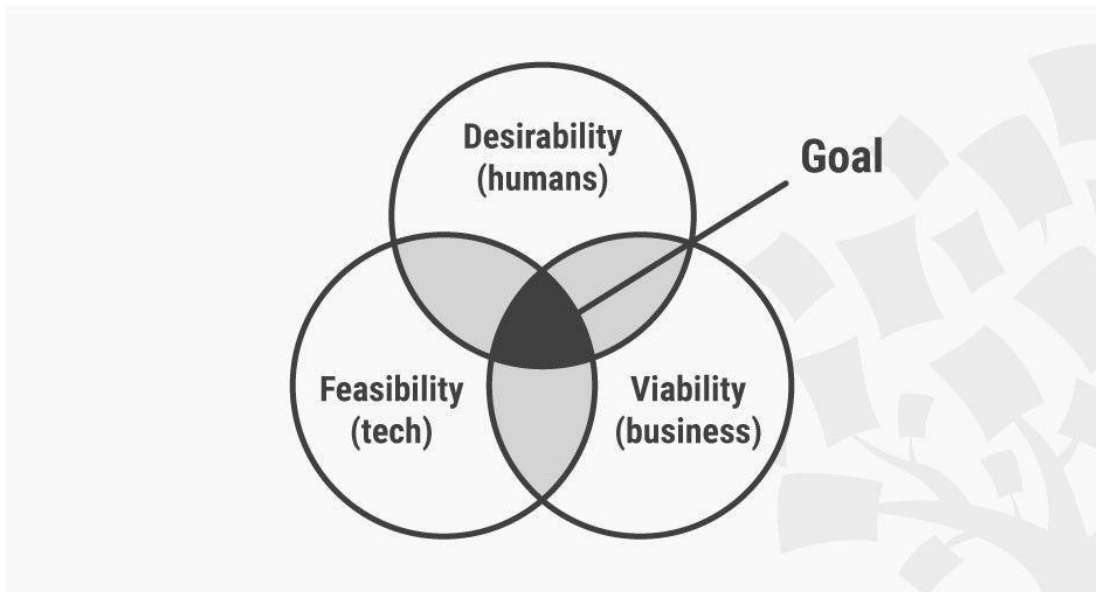
Mass consumption, driven by the industrial revolution and the invention of factories, is putting a huge environmental cost on our planet.

What They Say and What They Don't Say

People do not always convey all the details. They may withhold information out of fear, distrust or some other inhibiting factor, be it internal or based on those with whom they are engaging. Additionally, they may express themselves in ways not extremely articulate, thus requiring the listener to make sense of what is not being said or what is being hinted at, beneath the external expressions and words. As designers, we need to develop intuition, imagination, emotional sensitivity, and creativity in order to dig deeper without prying too personally, in order to extract the right kinds of insight so as to make a more meaningful difference. In other words, we need empathy so as to understand people thoroughly. Empathy is the difference between taking what your users say at face value and observing what IDEO Executive Design Director Jane Fulton Suri describes as “thoughtless acts” — small acts people exhibit that reveal how their behaviours are shaped by their environments. When people perform thoughtless acts such as hanging their sunglasses on their shirts, or wrapping coloured stickers around their keys to differentiate them, it’s a sign of how an imperfectly tailored environment forces an almost unconscious reaction on their part. However, we can find opportunities for new insights and new solutions to help people within unconscious acts.

EMPATHY IS CRUCIAL TO BUSINESS SUCCESS

Many leaders within the innovation, learn, and entrepreneurship spaces in which Design Thinking is prevalent have repeatedly pointed to three key parameters that define a successful product or service. They are: desirability, feasibility, and viability.



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It is not enough that the technology or means exist (i.e., feasibility is present) and that profits or business benefits may be derived (i.e., it is viable). It is essential for users to actually have a sense of desirability towards a solution. We can only fully understand and design a desirable product or service when people's needs, experiences, wants, and preferences are properly understood.

From a purely business profit-driven perspective, empathy is an essential component of any sound business solution. If we develop solutions in isolation, without essential insights about our users, we may create solutions that completely miss the mark and thus be ignored by the market. For example, many MP3 players have come and gone without much creating much of an impact, whereas the iPod was very successful at not only providing a technological solution but also providing a completely desirable and profitable experience, which resulted in Apple's taking a market lead.

As Frank Chimero, illustrator and author of *The Shape of Design*, says:
“People ignore design that ignores people”.
– Frank Chimero

DESIGNING WITHOUT EMPATHY: GOOGLE GLASS

Google launched its first wearable product, the Google Glass, with much fanfare in 2013. The head-mounted wearable computer, while being technologically impressive, failed to perform well, and a lot of that comes down to a lack of empathy towards the users.

Although the Glass allows users to take photos, send messages and view other information such as weather and transport directions, it does not actually fulfil the real needs of users. In other words, although the Glass performs many things, these are not things you need or want to get done.

Also, the Glass is generally a voice-activated device, and in our current social environment, saying commands out loud in the streets such as, “Okay Glass, send a message,” just isn't a socially acceptable thing to do. Google's lack of empathic understanding in the user's social environment is evident here; if the user has to perform socially awkward or unacceptable acts to be able to use your product, you can be sure that few people would be willing to use your product.

Lastly, the Glass featured a nondescript camera which resulted in privacy concerns for those people around the Glass user, since there was no way of knowing whether or not they were being filmed. All of these problems can be traced back to Google's lack of empathy when they designed the Glass, and this point is summed up nicely by the MIT Technology Review in one sentence:

“No one could understand why you'd want to have that thing on your face, in the way of normal social interaction.”
– MIT Technology Review



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The commercial failure of Google Glass can be traced to Google's lack of empathy towards users: voice-activated actions are socially awkward, the camera creates a privacy concern for people around the Glass user, and the device doesn't seem to solve any specific [user needs](#).

SUCCESS WITH EMPATHY: THE EMBRACE WARMER

With empathy, we can gain insights that could not be gathered by any other methods short of highly accurate calculated guesses. A team of postgraduate students at Stanford were tasked with developing a new [type](#) of incubator for developing countries. Their direct contact with mothers in remote village settings who were unable to reach hospitals, helped them to reframe their challenge to a warming device rather than a new kind of incubator.



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The end result was The Embrace Warmer, which has the potential to save thousands of lives. The Embrace Warmer is capable of going where no incubator could go before, due to its portability and dramatically reduced production costs. The Embrace Warmer is an ultra-portable incubator which can be

wrapped around an infant and be used while the infant is held in the mother's arm. Instead of needing to deposit their babies into far-flung hospitals, mothers in remote villages can use a portable warmer that serves the same need instead.

Had the team only thought of designing incubators, they may have developed a semi-portable lower cost incubator, which would still not have made it into remote villages. However, with the help of empathy—i.e., understanding the problems mothers in remote villages face—the design team designed a human-centered solution that proved to be optimal for mothers in developing countries. The objective of empathic research is uncovering, at times, intangible needs and feelings, that indicate what should ideally change in the product, system, or environment we're focusing on. Empathic research reveals the deeper needs and root causes, which, if addressed correctly, may profoundly change the project we're investigating. Instead of constantly designing new patches to cover or ease the symptoms only momentarily, we have the power to create a paradigm shift and provide a wide range of benefits packaged into a single solution. We can create new markets and move whole communities closer to higher order needs and goals. We can change the world when we operate at the appropriate levels.

ANYONE CAN MASTER EMPATHY

The empathy aspects of Design Thinking are named differently depending on whose version you might be following, but the core is essentially the same—i.e., being deeply human-centered. Different schools and Design Thinking companies have called empathic research "the Empathise stage" (which is the term we use), "the understand phase", and "the hear phase", and "looking", as well as a number of other terms.

If you are worried that you are unable to master the ability to be empathetic towards the people you are designing for fully, there is good news. Neuroscientists have recently discovered that empathy is hard-wired into the way humans are made and is an integral part of our physiology. They discovered that while humans observe others performing certain actions, or experience certain states, the observer's brain activity resembles someone actually engaged in the activity being observed. In other words, empathy is an innate quality that we can all make use of in order to design for the people around us.

We have all experienced the flurry of emotions or the rush of adrenalin experienced though merely observing someone else engaged in certain activities. We are empathic beings by our very nature, though, to a large extent, our social contexts and learning may work to remove this built-in empathy, or, at the very least, tame it. When you engage with the people you design for, keeping an open mind and being conscientious about developing empathy is key to a successful Design Thinking process and end product.

THE TAKE AWAY

Empathy is important for us as designers and particularly for design thinkers because it allows us to truly understand and uncover the latent needs and emotions of the people we are designing for. As such, we can design solutions that meet the three parameters of a successful product or service: desirability, feasibility and viability. In Design Thinking, we call this "the Empathise stage". Designing with empathy is what separates a human-centred product like The Embrace Warmer from another, such as Google's Glass. The

good news is that everybody can master empathy and become a great design thinker: we are all innately empathic.

COLLABORATION IN DISTRIBUTED DESIGN

Below are some reflections and ideas on fostering a thriving design organisation within a globally distributed product team — and how to leverage the best of asynchronous and real-time communication.

Starting in 2016, [Whisk](#) evolved into a distributed company and I started working less and less in a physical office space with team members around me. While I greatly value the freedom and flexibility that distributed working offers, I do miss the magic of in-person design collaboration. Being distributed naturally introduces friction and forces us to adopt new ways of working together.

Since Whisk went fully distributed, I've always been experimenting with ways to create a digital version of the 'open office' of yesteryear, where you can stroll over to another designer's desk to see what they are working on, using the technology at our disposal and the benefits of asynchronous collaboration.



How might we recreate in-person collaboration within digital environments?

Here are a few tactics that we've used in our globally distributed design team

1. ENSURE ALL WORK IS TRANSPARENT AND ACCESSIBLE

This might sound like a no-brainer: when designers are spread across the world working within their own documents/files, collaboration is filled with friction. Google Docs default to private. Figma files end up in your personal (not team) project. No one can lean over your shoulder and take a peek at your work. It takes a conscious effort to make sure ideas, documents, designs, etc... are easily accessible for others to jump in, see progress, and contribute.

To achieve this kind of open digital studio you need two things:

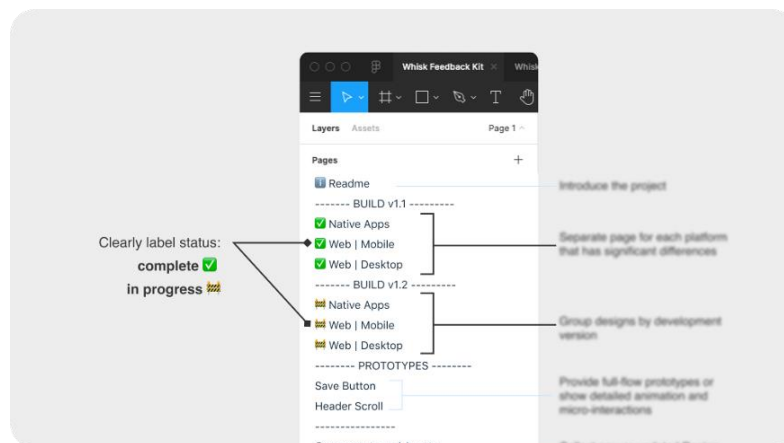
1. A culture that deeply understands and appreciates early collaboration. To be truly open you need to remove the stigma of “not ready to share” work. The team needs to be willing to show work at early stages. And it’s crucial to be sensitive in nurturing early ideas. Trust within the team plays an important role.
2. Systems and processes that default to shared and accessible for all those within the organization. Creating a new Figma design file should end up in a team-accessible project and be cross-linked to Notion/GoogleDocs/etc so others immediately know where to find it.

2. CLEARLY DEFINE STANDARDS AND PROCESS AROUND FILE ORGANIZATION

After defaulting to transparent and accessible, the next step is aligning the team around how to organize design work. Creativity is a naturally messy process, but a guiding framework for how Figma files are setup makes it easier for other designers to dive in, navigate, contribute, and borrow ideas for their own projects. It makes all our work modular, interconnected, and intuitive to work with.

Some of the processes that help us stay aligned:

- A readme page — every file needs to have a readme page that sets the context (objective) of the project, shows it’s status, who owns it, and links to other relevant documents (such as Notion for a detailed PRD).
- Project folders — putting your file in the right project means that, by default, the right people have access to it. Simple!
- Shared style and component libraries — all our work is connected to common style and pattern libraries so we’re always working from the same source of truth: our design system.
- Common page and layer structures — conventions around page organisation makes navigating files easier. Designers know where to drop their feedback and engineers know where to find the production-ready designs.



Standardising design file organisation helps everyone navigate and contribute.

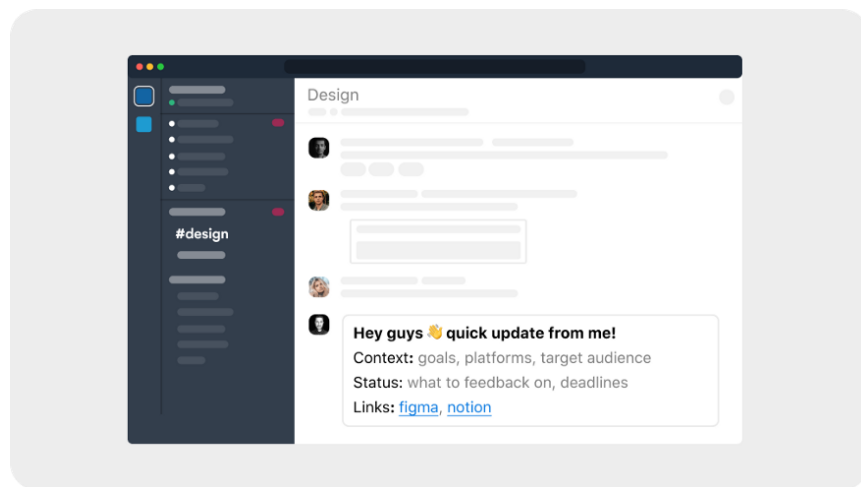
3. Develop habitual asynchronous communication

Making all work open and accessible by default reduces some of the typical “communication” needed to collaborate, but building a strong muscle for quick, async communication is even more powerful (and necessary) to have strong collaboration between team members spread across timezones.

Through thoughtful and timely async communication, our design engine keeps projects moving day and night — ideas are constantly being built upon and improved by team members around the globe.

Some ways we’re doing this:

Slack updates — designers share short updates that signal progress and focuses attention on a specific area of the work. A quick message a few times per week with some context + links to design goes a long way to having an open and collaborative culture.



We use Slack for ongoing asynchronous design updates and feedback cycles.

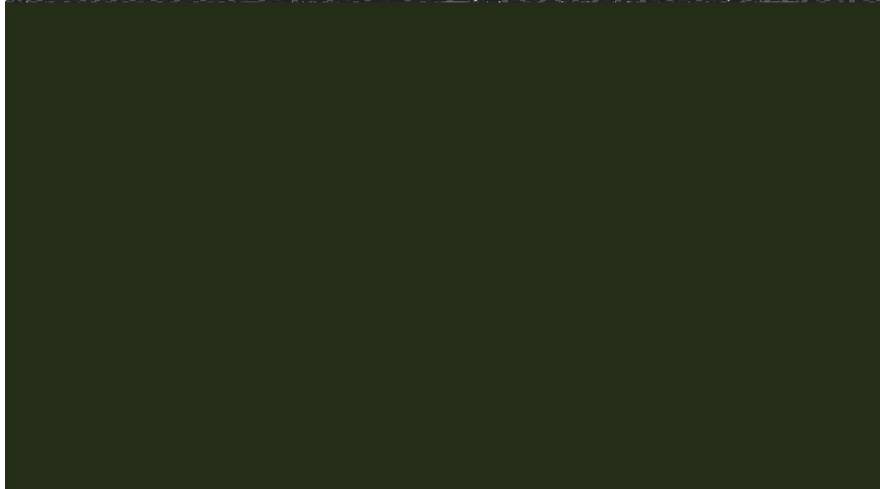
Showing thought process in Figma —capturing design thinking next to the designs helps others get up to speed faster and give more valuable feedback. Give context on constraints, directions explored, and where you really need feedback.



Capturing design process within Figma is a powerful way to work asynchronously.

Commenting within Figma —using the built-in comment feature is the standard/easy way to share ideas and feedback. It works really well because it's in context (right on top of the designs) and actionable (it's can be treated as a to-do).

Video messages — showing and speaking to the work is the highest bandwidth way to transfer ideas. We use [Loom.com](https://loom.com). We try to keep our video messages to less than 5 minutes (and of course encourage liberal use of the 2x playback option).



Loom, iPad, Pencil, and Sidecar are a powerful combination for fast visual feedback.

4. DEFAULT TO VISUAL FEEDBACK (THE DESIGNER'S SUPERPOWER)

All the above methods have a time and place, but the most effective tactic we use across all of them is visual feedback.

Instead of writing or explaining feedback we strive to show ideas (in any medium: from crude google slides boxes-and-arrows to ipad sketches to high-fidelity prototypes). Anything other than words will get the idea across more quickly and more clearly. Defaulting to visual rather than written or spoken feedback catalyses the entire design and development process. It also immediately reveals if it's bad feedback (for example: it looks worse, it can't be executed, it's too half-baked).



Sometimes crude tablet sketches are the best way to nurture and spread ideas.

This is an often under-utilised superpower of designers and helps supercharge every day situations, it helps other designers take the next step faster. Learning to communicate visually quickly is something we're actively training within the design team here.

Figma makes visual ideation so easy with component libraries. It's easy to create prototypes at lightning speed. But high-fidelity isn't always appropriate or accessible so another tactic I've been exploring recently (especially for ideas that are more early stage and abstract), is using my iPad and Pencil to create "napkin" sketches in real-time. It's surprisingly fun, creative, and collaborative on otherwise mundane video calls.

5. MAINTAIN REGULAR REAL-TIME COLLABORATION MOMENTS

Great design work cannot sustain asynchronous communication alone. That's why we do weekly Design Syncs (a quick round robin of updates and peeks into the work) and Design Reviews (deep dive discussions into the user experience and visual design work). These moments allow us to have real-time discussions about the work, but also give us the chance to enjoy fun/random/personal discussions! I've seen it build trust and relationships within the team and bring energy to the work, even if the discussions themselves don't always result in the best ideas.